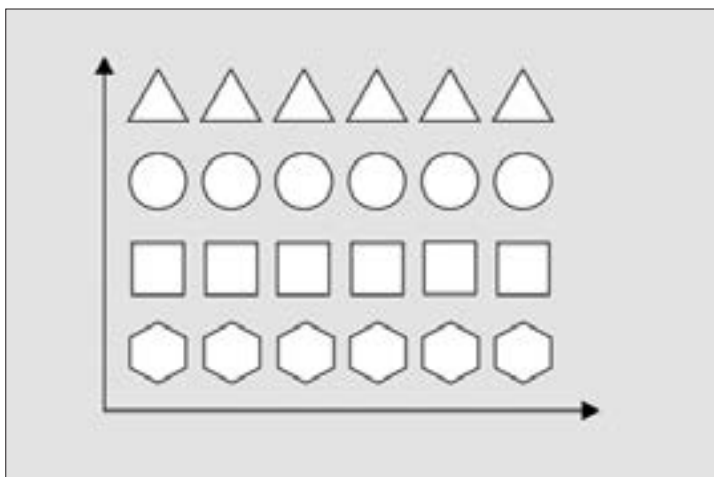




The four conversations we shall start with

Frequency Division Multiple Access - FDMA

FDMA is the oldest and most basic method for multiple access on a cellular network. It was used for analogue transmission on the first generation of cellular networks. Though it was used for analogue, it was perfectly capable of digital transmission as well; it just wasn't considered an efficient way to do so.

Each cellular operator would be authorised to use a range of frequencies. The operator would then divide this range into equal-sized frequency bands, taking into consideration two things: first, each frequency band had to be large enough to accommodate the frequencies of the human voice. Keeping the band 30 kHz wide seemed to be sufficient. Second, they needed to ensure that two calls on the network did not interfere with each other. To do this, there would be a 'guard band' between the frequencies that would actually be used for calls.

Each call is assigned a pair of frequencies - one for talking and one for listening. Once the call is done, this pair is released and available to be re-used for another call.